

BUS123 · MATH-M11-L01 · Pre-Reading

READING AND ANALYZING FINANCIAL REPORTS

Harborside Medical Center · Fall 2026

Connect to Prior Knowledge

In MATH-M10 we explored stocks and bonds — tools investors use to put capital *into* a business. Now we flip the lens entirely. Before any investor decides to buy a bond or a share of stock, they read the company's financial reports. This lesson teaches you to read those reports yourself. Whether you're evaluating a potential employer, managing a department, or eventually running your own organization, the three financial statements we cover today are the most important documents in business.

PART 1 · THE ACCOUNTING EQUATION AND THE BALANCE SHEET

The Foundation: Assets = Liabilities + Equity

Every financial report ever produced is built on a single mathematical identity:

Formula: Assets = Liabilities + Owner's Equity

This is the **Fundamental Accounting Equation**, and it never breaks. Think of it as a perfectly balanced scale. The left side represents everything a business *owns or controls*. The right side represents every *claim* against those things — either from outside lenders (liabilities) or from the owners themselves (equity).

Because of **double-entry accounting**, every transaction touches at least two sides of the equation to keep it balanced. If Harborside Medical Center buys a new MRI machine for \$400,000, one of two things happens:

- If they pay cash: Assets go down \$400K (cash) and up \$400K (equipment) — a wash. Equation stays balanced.
- If they borrow: Assets go up \$400K (equipment) and Liabilities go up \$400K (loan). Equation stays balanced.

There is no transaction in the history of any business that breaks this rule.

The Balance Sheet

The **Balance Sheet** is the formal statement that presents the accounting equation as a snapshot at a specific point in time — usually December 31. It has three sections:

Assets — what the business owns:

- *Current assets*: Cash, Accounts Receivable (money owed by patients/customers), Inventory, Prepaid Expenses. These convert to cash within one year.

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MATH-M11-L01

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PRE-READING

- *Non-current assets*: Buildings, Equipment, Vehicles. These are used over multiple years. The portion of their cost allocated to each year is called **Depreciation**.

Liabilities — what the business owes to external parties:

- *Current liabilities*: Accounts Payable, Salaries Payable, current loan installments — due within one year.
- *Long-term liabilities*: Bank loans, bonds — due after one year.

Owner's Equity — the residual value left for owners after all debts are paid: **Formula:** Owner's Equity = Assets – Liabilities Equity grows through two engines: owner contributions and **Retained Earnings** (profits kept in the business rather than paid out).

Note

Harborside Example: At year-end, Harborside had total assets of \$3,027,000, total liabilities of \$1,305,000, and owner's equity of \$1,722,000. Verify: $\$1,305,000 + \$1,722,000 = \$3,027,000$ ✓

PART 2 · THE INCOME STATEMENT

Did the Business Make Money?

While the balance sheet is a *snapshot*, the **Income Statement** is a *motion picture* — it covers a period of time (usually a full year) and tells the story of whether the business was profitable.

The basic structure for a **service business** like a hospital:

Income Statement structure:

- Net Revenue (Patient Services)
- – Cost of Services
- = Gross Profit
- – Operating Expenses
- = Net Income

For a **retail business**, the COGS calculation includes inventory:

Formula: $\text{COGS} = \text{Beginning Inventory} + \text{Net Purchases} - \text{Ending Inventory}$

Net Sales is calculated after removing returns and discounts:

Formula: $\text{Net Sales} = \text{Gross Sales} - \text{Sales Returns} - \text{Sales Discounts}$

Note

Harborside Example: Gross patient service revenue was \$4,680,000. After contractual adjustments of \$430,000 (insurance discounts), net revenue = \$4,250,000. After subtracting cost of services (\$2,285,000) and operating

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MATH-M11-L01

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PRE-READING

expenses (\$1,420,000), Harborside's net income = \$545,000.

PART 3 · THE STATEMENT OF CASH FLOWS

"Cash is King" — Why Profitable Businesses Can Still Fail

Here is one of the most important business lessons you will ever learn: **profit and cash are not the same thing.**

The Income Statement operates on the *accrual basis*: revenue is recorded when *earned*, expenses when *incurred* — regardless of when actual cash moves. A hospital can bill \$680,000 in patient services in December, record that as revenue, and show a healthy profit on paper — but if the insurance companies don't pay until February, that cash doesn't exist yet. Meanwhile, staff payroll is due Friday.

The **Statement of Cash Flows** fixes this. It tracks only real dollar movements and organizes them into three categories:

Section	What It Covers	Harborside Example
Operating Activities	Daily business: collecting payments, paying suppliers and staff	+\$310,000
Investing Activities	Long-term asset purchases/sales: buying equipment, selling old assets	–\$580,000
Financing Activities	Loans and owner transactions: borrowing, repaying, dividends	+\$220,000

The net change in cash reconciles the opening and closing cash balances on the Balance Sheet. For Harborside: \$462,000 opening + \$310,000 – \$580,000 + \$220,000 = **\$412,000 closing cash** — which must match the cash line on the Balance Sheet. ✓

PART 4 · TREND ANALYSIS AND RATIO ANALYSIS

Trend Analysis: Watching the Direction

A single year's numbers don't mean much without context. **Trend Analysis** solves this by expressing each year's amount as a percentage of a *base year*.

Formula: Trend % = Current Year Amount ÷ Base Year Amount

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MATH-M11-L01

BUS 123 — MATH-M11-L01 — Reading and Analyzing...

PRE-READING

The base year is always 100%. Every other year shows whether performance improved or declined relative to that starting point.

Note

Harborside Example: Using 2024 as the base year (100%), net revenue grew to 112% in 2026 — a 12% improvement. But net income grew from 100% to 141% — meaning Harborside is getting *more efficient*, not just bigger.

Ratio Analysis: Measuring Performance

Ratio Analysis uses division formulas to produce standardized metrics that can be compared across years and against industry benchmarks. Six key ratios for this lesson:

Ratio	Formula	Industry Benchmark
Current Ratio	Current Assets ÷ Current Liabilities	2.0 : 1
Quick Ratio	(Cash + AR) ÷ Current Liabilities	1.0 : 1
Total Debt to Total Assets	Total Liabilities ÷ Total Assets	50%–70%
Return on Equity (ROE)	Net Income ÷ Owner's Equity	15%–20%
Profit Margin on Net Sales	Net Income ÷ Net Sales	8%–12% (healthcare)
Asset Turnover	Net Sales ÷ Total Assets	Varies by industry

Note

Important: A ratio only becomes meaningful when compared to something — prior year results, industry averages, or a competitor's numbers. A current ratio of 1.5 could be perfectly healthy in one industry and a warning sign in another.

CHECK YOUR UNDERSTANDING

Answer these five questions before class. Refer back to the sections above — these are designed to confirm you've absorbed the key ideas.

1. Harborside Medical Center has total assets of \$3,027,000. If total liabilities are \$1,305,000, what is the owner's equity? Show your calculation.
2. Harborside's gross patient service revenue was \$4,680,000 and contractual adjustments were \$430,000. What is net patient service revenue? Which statement does this appear on?

BUS 123 - Solving Business Problems with Technology

MATH-M11-L01

BUS 123 — MATH-M11-L01 — Reading and Analyzing...

PRE-READING

3. A hospital reports net income of \$600,000 but its cash flow from operations is only \$85,000. Give one reason why this gap could exist.
4. Using 2024 as the base year, Harborside's net income was \$320,000 in 2024 and \$450,000 in 2026. What is the trend percentage for 2026? Round to the nearest whole percent.
5. Harborside's current ratio is 2.43. The industry average is 2.0. What does this tell you about Harborside's ability to pay its short-term obligations?

KEY VOCABULARY

Term	Definition
Accounting Equation	Assets = Liabilities + Owner's Equity — the fundamental identity that governs all financial reporting
Balance Sheet	A financial snapshot showing what a business owns, owes, and is worth at a specific date
Accrual Basis	Revenue recognized when earned, expenses when incurred — regardless of cash timing
Accounts Receivable	Money owed to the business by customers for services already delivered
Depreciation	Spreading the cost of a long-term asset over its useful life
Income Statement	Reports revenues, expenses, and net income over a period of time
Cost of Goods Sold (COGS)	The direct cost of producing what was sold (Beginning Inventory + Purchases – Ending Inventory)
Gross Profit	Net Sales minus Cost of Goods Sold
Net Income	The bottom-line profit after all revenues and expenses are accounted for
Statement of Cash Flows	Tracks actual cash inflows and outflows across three activity types

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MATH-M11-L01

BUS 123 — MATH-M11-L01 — Reading and Analyzing...

PRE-READING

Term	Definition
Operating Activities	Cash flows from the day-to-day revenue-generating activities of the business
Retained Earnings	Accumulated profits kept in the business rather than paid out to owners
Trend Analysis	Expressing each year's financial data as a percentage of a designated base year
Ratio Analysis	Using division formulas to create standardized financial metrics for comparison
Current Ratio	$\text{Current Assets} \div \text{Current Liabilities}$ — measures short-term liquidity
Return on Equity	$\text{Net Income} \div \text{Owner's Equity}$ — measures how effectively equity generates profit